

## Differential Equations:

DE:  $\rightarrow$  1/more dependent variables

$\rightarrow$  1/more independent variables.

$$\begin{array}{l} \frac{dy}{dx} \rightarrow \text{dependent} \\ \frac{dy}{dx} \rightarrow \text{independent} \end{array}$$

\* the one which is differentiated is dependent variable

\* the one which is w.r.t. to sth is differentiated is independent variable

$\Rightarrow$  An eq<sup>n</sup> involving derivatives of one/more dependent variables w.r.t one/more independent variables is called a differential equation.

$$\frac{d^2y}{dx^2} + xy \left( \frac{dy}{dx} \right)^2 = 0 \longrightarrow \text{ODE}$$

$$\frac{d^4 x}{dt^4} + 5 \frac{d^2 x}{dt^2} + 3x = \sin t \longrightarrow \text{ODE}$$

$$\frac{dv}{du} + \frac{dv}{dt} = v \longrightarrow \text{PDE}$$

$$\frac{d^2 u}{dx^2} + \frac{dx^2}{dy^2} + \frac{d^2 u}{dz^2} = 0 \longrightarrow \text{PDE}$$

$$\frac{d}{dx}(e^{ax}) = ae^{ax}$$

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

 $\longrightarrow \text{Identities}$

\* DE

- $\longrightarrow$  ODE (Ordinary DE)  $\longrightarrow$  1 independent variable
- $\longrightarrow$  PDE (Partial DE)  $\longrightarrow$  More than 1 independent variable

Order  $\Rightarrow$  Highest no. of times a dependent variable is differentiated.

Degree  $\Rightarrow$  Power of a derivative.

$$\left(\frac{dy}{dx}\right)^2 + 6 - 5x = 0$$

order : 1  
degree : 2

\*ODE has two types:

└→ Linear (Degree = 1)  
└→ Non linear (Degree > 1)

①  $\frac{d^4 y}{dx^4} + 5y = 0 \rightarrow \text{Linear.}$

②  $x^4 \left( \frac{d^3 y}{dx^3} \right) + x^3 \left( \frac{d^2 y}{dx^2} \right) + 3 = 0 \rightarrow \text{Non linear}$

③  $\left( \frac{dy}{dx} \right)^4 + 6 = 0 \rightarrow \text{Non-linear}$

\*\*\* ④  $\frac{d^4 y}{dx^4} + 5y^2 = 0 \rightarrow \text{Non-linear}$   
└→ power of y more than 1

⇒ Newton's Law: (ODE)

$$F = ma$$
$$m \frac{d^2 x}{dt^2}(t) = F\left(t, x(t), \frac{dx}{dt}(t)\right)$$

↓  
depends on t

↑  
position

↓  
velocity

$$\begin{bmatrix} \text{Order} = 2 \\ \text{degree} = 1 \end{bmatrix}$$

## ⇒ Radioactive Decay (ODE)

$u \rightarrow$  amount of radioactive particles

$$\frac{du}{dt}(t) = -ku(t) \quad \left[ \begin{array}{l} \text{order} = 1 \\ \text{degree} = 1 \end{array} \right]$$

## ⇒ Heat (PDE)

$T \rightarrow$  temperature,

$x(x, y, z) \rightarrow$  space

$$\frac{dT}{dt}(t, x) = k \left( \frac{d^2T}{dx^2}(t, x) + \frac{d^2T}{dy^2}(t, x) + \frac{d^2T}{dz^2}(t, x) \right)$$

$$\left[ \begin{array}{l} \text{order} = 2 \\ \text{degree} = 1 \end{array} \right]$$

$$y' = ay + b$$

constant coefficients.

$$y' = -\frac{2}{t}y + 4t$$

variable coefficients.

(MIEHL  
GAP)\* Ordinary Differential Equation:

→ Determine actually what function/functions satisfies the differential equation.

$$\boxed{2x+y=5} \quad (0, 5), (1, 3)$$

Q. Which of the following is a solution to,

$$\frac{d^2y}{dx^2} = -4y$$

(i)  $\cos(4x) = y$ .

(ii)  $y = \sin 2x$ .

~~$y = \cos 2x$~~

(ii)  $y = \sin 2x$

$$\Rightarrow \frac{dy}{dx} = 2 \cos 2x$$

$$\Rightarrow \frac{d^2y}{dx^2} = -4 \sin 2x$$

$$\therefore \frac{d^2y}{dx^2} = -4y$$

(i)  $\cos 4x = y$

$$\Rightarrow \frac{dy}{dx} = -4 \sin 4x$$

$$\Rightarrow \frac{d^2y}{dx^2} = -16y$$

Q. 11.3  $y' = 2y + 3$ .

Given,  $y = e^{2t} - 3/2$   
 $\Rightarrow y' = 2e^{2t}$  (LHS).

RHS,  $= 2y + 3$ .

$$= 2(e^{2t} - 3/2) + 3$$

$$= 2e^{2t} - 3 + 3$$

$$= 2e^{2t} = \text{LHS}$$

Shown

$$1.1.4: y = 4e^{2t} + 3.$$

$$\Rightarrow \frac{dy}{dx} = 8e^{2t}.$$

$$\Rightarrow \frac{dy}{dx} = 2.$$

$$1.1.4: y = 4e^{2t} + 3$$

$$\Rightarrow \frac{dy}{dt} = 8e^{2t}$$

$$\Rightarrow \frac{dy}{dt} = 2(4e^{2t} + 3) - 3$$

$$\Rightarrow y' = 2y - 3$$

Constant Coefficient:

→ General solution of DE.

General solution for the ODE's of the form  $y' = ay + b$ . (constant coeff.)

Th<sup>m</sup>:  $y' = ay + b$  has infinitely many solutions of the form

$$y(t) = ce^{at} - \frac{b}{a}, \quad c \in \mathbb{R}$$

Proof:-

1. When  $b=0$ ,

$$y' = ay.$$

$$\Rightarrow \frac{y'}{y} = a$$

$$\Rightarrow \frac{d}{dt}(\ln|y|) = a$$

$$\Rightarrow (\ln|y|)' = a$$

$$\Rightarrow \int (\ln|y|)' = \int a \quad [\text{Integrating w.r.t } t]$$

$$\Rightarrow \ln|y| = at + c_0.$$

$$\Rightarrow y = \pm e^{at+c_0}.$$

$$\Rightarrow y = e^{c_0} \cdot e^{at}.$$

$$\Rightarrow \boxed{y = c \cdot e^{at}.$$

2. When  $b \neq 0$ ,

$$y' = ay + b.$$

$$\Rightarrow y' = a(y + \frac{b}{a})$$

$$\Rightarrow y' + (\frac{b}{a})' = a(y + \frac{b}{a}).$$

$$\Rightarrow (y + \frac{b}{a})' = a(y + \frac{b}{a}).$$

$$\Rightarrow \frac{(y + \frac{b}{a})'}{y + \frac{b}{a}} = a$$

$$\Rightarrow (\ln|y + \frac{b}{a}|)' = a.$$

$$\Rightarrow \ln|y + \frac{b}{a}| = at + c_0$$

$$\Rightarrow y + \frac{b}{a} = \pm e^{at+c_0}$$

$$\Rightarrow y + \frac{b}{a} = ce^{at}$$

$$\boxed{\therefore y = ce^{at} - \frac{b}{a}}$$

1.1.5:  $y' = 2y + 3.$

$$\Rightarrow y' = 2(y + \frac{3}{2}).$$

$$\Rightarrow \frac{(y + \frac{3}{2})'}{y + \frac{3}{2}} = 2.$$

$$\Rightarrow \int (\ln |y + \frac{3}{2}|)' = \int 2$$

$$\Rightarrow \ln |y + \frac{3}{2}| = 2t + c.$$

$$\Rightarrow y + \frac{3}{2} = e^{2t} \cdot e^c$$

$$\Rightarrow y = ce^{2t} - \frac{3}{2}$$

Ans

1.1.4

MATH-4241:

7/3/23

## \* Integrating Factor Method (IFM)

$y \rightarrow$  integrating factor.

$$y' = ay + b.$$

$$\Rightarrow y' - ay = b$$

$$\text{cloud: } -ay = y'$$

$$\Rightarrow y y' - y a y = y b.$$

$$\Rightarrow y y' + u' y = y b.$$

$$\Rightarrow (uy)' = y b.$$

$$-ay = y'$$

$$\Rightarrow \frac{y'}{y} = -a$$

$$\Rightarrow \int \ln|y| = \int -a \quad [\text{Integrating with } \pm]$$

$$\Rightarrow \ln|y| = -at + c.$$

$$\Rightarrow y = \pm e^{-at} \cdot e^c.$$

$$\Rightarrow y = c_0 \cdot e^{-at}$$

$$\Rightarrow \boxed{y(t) = e^{-at}}.$$

$$\begin{pmatrix} te = c_1 \\ c_1 = 1 \end{pmatrix}$$

From (i)  $\Rightarrow$

$$(e^{-at}y)' = (be^{-at}) \frac{d}{dt} \left( -\frac{b}{a} e^{-at} \right).$$

$$\Rightarrow (e^{-at}y)' = \left( -\frac{b}{a} e^{-at} \right)'$$

$$\Rightarrow \left( e^{-at}y + \frac{b}{a} e^{-at} \right)' = 0$$

$$\Rightarrow e^{-at}y + \frac{b}{a} e^{-at} = c$$

$$\Rightarrow y + \frac{b}{a} = ce^{at}$$

$$\Rightarrow \boxed{y(t) = ce^{at} - \frac{b}{a}}$$

Q. Find the unique solution of the problem:

method  $y' = -3y + 1$  by integrating factor  
 $y(0) = 1$ , given.

$$y' = -3y + 1. \quad \text{--- (i)}$$

$$\mu = e^{-at} = e^{3t}.$$

Multiplying  $\mu = e^{3t}$  in (i)  $\Rightarrow$

$$e^{3t}y' = -3e^{3t}y + e^{3t}$$

$$\Rightarrow e^{3t}y' + 3e^{3t}y = e^{3t}$$

$$\Rightarrow e^{3t}y + (e^{3t})'y = e^{3t}$$

$$\Rightarrow \int (e^{3t}y)' = \int e^{3t}$$

$$\Rightarrow e^{3t}y = \frac{1}{3}e^{3t} + c$$

$$\Rightarrow y = ce^{-3t} + \frac{1}{3} \quad [\text{Multiplied by } e^{-3t}]$$

$$\Rightarrow 1 = ce^0 + \frac{1}{3} \quad [y(0) = 1]$$

$$\therefore c = \frac{2}{3}$$

$$y(t) = \frac{2}{3}e^{-3t} + \frac{1}{3}$$

Ans

## \* Linear Variable Coefficient:

$$y' = a(t)y + b(t)$$

~~$y' = a(t)y + b(t)$~~   $y' - ay = b$ .  $[a(t) = a, b(t) = b]$

~~$y' = a(t)y + b(t)$~~

$$\Rightarrow \mu y' - a\mu y = \mu b$$

$$\Rightarrow \mu y' + y' \mu = \mu b$$

~~$y' = a(t)y + b(t)$~~

$$\mu = e^{-A(t)} \quad A(t) = \int a(t) dt$$

$$\mu' = -a\mu$$

$$\Rightarrow \int (e^{-A} y)' = \int e^{-A} b$$

$$\Rightarrow e^{-A} y = \int e^{-A} b + c$$

$$\Rightarrow e^{-A} y = \int e^{-A} b + c \quad [\text{Multiplied with } e^A]$$

$$\Rightarrow y = e^A \int e^{-A} b + ce^A$$

$$\Rightarrow y(t) = ce^A + e^A \int e^{-A} b$$

$$\Rightarrow y(t) = ce^{A(t)} + e^{A(t)} \int e^{-A(t)} b(t) dt$$

Proved

~~Find all solutions to the DE.~~

$$y' = \frac{3}{t}y + t^5$$

→

Solution:

$$y' = \frac{3}{t}y + t^5$$
$$\Rightarrow y' - \frac{3}{t}y = t^5$$

Finding value of  $\mu$

$$-\mu\mu' = \mu'$$

$$\Rightarrow \frac{\mu'}{\mu} = -\mu = -\frac{3}{t}$$

$$\Rightarrow \int (\ln|\mu|)' = \int -\frac{3}{t}$$

$$\Rightarrow \ln|\mu| = -3 \ln|t| + C_0$$

$$\Rightarrow \ln|\mu| = \ln|t^{-3}| + C_0$$

$$\Rightarrow \mu = \pm e^{\ln(t^{-3})} \cdot e^{C_0}$$

$$\Rightarrow \mu = t^{-3}$$

*[Signature]*

$$y' = \frac{3}{t}y - t^5$$

$$\Rightarrow ty' = \frac{3}{t}ty = t^5$$

$$\Rightarrow t^{-3}y' = \frac{3}{t} \cdot t^{-3}y = t^5 \cdot t^{-3}$$

$$\Rightarrow t^{-3}y' = 3t^{-4}y = t^2$$

$$\Rightarrow t^{-3}y' = (t^{-3})'y = t^2$$

$$\Rightarrow (t^{-3}y)' = t^2$$

$$\Rightarrow (t^{-3}y)' = t^2$$

$$\Rightarrow \left(t^{-3}y - \frac{t^3}{3}\right)' = 0$$

$$\Rightarrow t^{-3}y - \frac{t^3}{3} = c$$

$$\boxed{y = ct^3 + \frac{t^6}{3}}$$

## 1.3: Separable Differential Equation: (SDE)

$$h(y), y' = g(t)$$

$$y' = \frac{t^2}{1-y^2}$$

$$\Rightarrow (1-y^2) y' = g(t). \rightarrow \text{SDE}$$

$$y' = e^y + \cos(t)$$

$$\Rightarrow y' - e^y = \cos(t)$$

$$\Rightarrow y' - h(y) = g(t) \rightarrow \text{not SDE}$$

$$y' = ay + b.$$

$$\Rightarrow \frac{y'}{ay+b} = 1.$$

$$\Rightarrow \frac{1}{ay+b} \cdot y' = 1.$$

$$\Rightarrow h(y) \cdot y' = g(t) \rightarrow \text{SDE}$$

$$h(y) = \frac{1}{ay+b}, \quad g(t) = 1$$

$$-\frac{y'}{y^2} = \cos(2t)$$

$$h(y) = \int -\frac{1}{y^2}.$$

$$g(t) = \int \cos 2t.$$

let  
 $u = y$   
 $\Rightarrow u' = y'$   
 $\Rightarrow \frac{dy}{dt} = \frac{du}{dt}$

$$\Rightarrow \int \frac{-y'}{y^2} dt = \int \cos 2t dt$$

$$\Rightarrow \int \frac{du}{u^2} = \frac{\sin 2t}{2} + c$$

$$\Rightarrow -\int u^{-2} du = \frac{1}{2} \sin 2t + c$$

$$\Rightarrow \frac{1}{u} = \frac{1}{2} \sin 2t + c$$

$$\Rightarrow \frac{1}{y(t)} = \frac{1}{2} \sin 2t + c$$

$$\frac{1}{y(t)}$$

$$y(t) = \frac{2}{\sin 2t} + 2c$$

Q. Find all solutions  $y$  to the DE  $y' = \frac{t^2}{1-y^2}$  ~~with~~  $y(0) = 1$ .

$$y'(1-y^2) = t^2$$

$$\begin{aligned} u &= y \\ \Rightarrow u' &= y' \end{aligned}$$

$$\Rightarrow \int y'(1-y^2) = \int t^2$$

$$\Rightarrow \int (1-y^2) du = \frac{t^3}{3} + c$$

$$\Rightarrow u - \frac{u^3}{3} = \frac{t^3}{3} + c$$

$$\Rightarrow y(t) - \frac{y(t)^3}{3} = \frac{t^3}{3} + c$$

$$\Rightarrow y(t) - \frac{y^3 t}{3} = \frac{t^3}{3} + \frac{2}{3}$$

Explicit Form:

$$\int h(y) y' = \int g(t) dt$$

$$\Rightarrow H(y(t)) = G(t) + c \quad \text{where } H(y(t)) = \int h(y) dy.$$

$$\Rightarrow y(t) = H^{-1}(G(t)) + c.$$

$$G(t) = \int g(t) dt$$



Q. Find the soln ~~of~~ in explicit form:

$$y' = \frac{2-t}{1+y}, \quad y(0) = 1$$

where,

$$h(y) = 1+y \rightarrow H(y) = y + \frac{y^2}{2}$$
$$g(t) = 2-t \rightarrow G(t) = 2t - \frac{t^2}{2}$$

$$y(t) + \frac{y^2(t)}{2} = 2t - \frac{t^2}{2} + C$$

$$\Rightarrow y(0) + \frac{y^2(0)}{2} = 0 + C$$

$$\Rightarrow 1 + \frac{1}{2} = C.$$

$$\Rightarrow C = \frac{3}{2}$$

$$y(t) + \frac{y^2(t)}{2} = 2t - \frac{t^2}{2} + \frac{3}{2}$$

$$\Rightarrow y^2(t) + 2y(t) - (t^2 - 4t + 3) = 0.$$

$$\therefore y_1(t) = \frac{-2 \pm \sqrt{4 - 4(t^2 - 4t + 3)}}{2}$$

$$= -1 \pm \sqrt{-t^2 + 4t + 4}$$

# Euler Homogeneous Equation:

1. Check Homogeneous or not.
2. Express the DE as  $F(y/t)$ .

Q. Show that  $(t-y)y' - 2y + 3t + \frac{y^2}{t} = 0$  is an Euler Homogeneous equation.

Soln: <sup>STEP-01</sup>  $(t-y)y' = 2y - 3t - \frac{y^2}{t}$

$$\Rightarrow y' = \frac{2y - 3t - \frac{y^2}{t}}{t-y}$$

$$\Rightarrow f(t,y) = \frac{2y - 3t - \frac{y^2}{t}}{t-y}$$

$$\Rightarrow f(ct, cy) = \frac{2cy - 3ct - \frac{c^2y^2}{ct}}{ct - cy}$$

$$\begin{aligned} \Rightarrow f(ct, cy) &= \frac{c(2y - 3t - \frac{y^2}{t})}{c(t-y)} \\ &= f(t,y) \end{aligned}$$

$\therefore$  Homogeneous

STEP 02.

$$\begin{aligned} f(t, y) &= \frac{(2y - 3t - \frac{y^2}{t}) \cdot \frac{1}{t}}{(1y) \cdot \frac{1}{t}} \\ &= \frac{2(\frac{y}{t}) - 3(\frac{y}{t}) - (\frac{y}{t})^2}{1 \cdot (\frac{y}{t}) - \frac{y}{t}} \\ &= F\left(\frac{y(t)}{t}\right) \end{aligned}$$

$$f(t, y) = F\left(\frac{y}{t}\right)$$

SHOWN

This is Euler homogeneous equation in standard form.

\* Bernoulli's Equation:

$$y' = p(t)y + q(t)y^n$$

Dividing by  $y^n$ :

$$\frac{y'}{y^n} = \frac{p(t)}{y^{n-1}} + q(t)$$

$$\rightarrow \frac{y'}{y^n} = p(t)y^{-(n-1)} + q(t) \quad \text{--- (i)}$$

\* Let,  $v = y^{-(n-1)}$ ; finding its derivative.

$$v' = [y^{-(n-1)}]'$$

$$\Rightarrow v' = -(n-1)y^{-n}y'$$

$$\Rightarrow -\frac{v'}{n-1} = \frac{y'}{y^n} \quad \text{--- (ii)}$$

From (i) and (ii)  $\Rightarrow$

$$-\frac{v'}{n-1} = \frac{p(t)y^{-(n-1)}}{y^n} + q(t)$$

$$\Rightarrow v' = -(n-1)p(t)v - (n-1)q(t)$$

which is the final form  
linear equation



$$Q. y' = y + 2y^5$$

$$\Rightarrow \frac{y'}{y^5} = \frac{y}{y^5} + 2 \quad [\text{Dividing by } y^5]$$

$$\Rightarrow \frac{y'}{y^5} = \frac{1}{y^4} + 2 \quad \text{--- (i)}$$

$$\text{Let, } v = \frac{1}{y^4}$$

$$\Rightarrow v' = (y^{-4})'$$

$$\Rightarrow v' = -4 \left( \frac{y'}{y^5} \right)$$

$$\Rightarrow \frac{-v'}{4} = \frac{y'}{y^5} \quad \text{--- (ii)}$$

From (i) and (ii), we can write:

$$\frac{-v'}{4} = \frac{1}{y^4} + 2$$

$$\Rightarrow \frac{v'}{4} = v + 2$$

$$\Rightarrow v' = 4v + 8 \quad (\text{L.E})$$

$$\Rightarrow v' + 4v = -8$$

Multiplying with IFM,  $\mu = e^{4t}$ .

$$\int (e^4 + v)' = \int -8e^{4t}$$

$$\Rightarrow e^{4t} v = -\frac{8}{4} e^{4t} + C$$

$$\Rightarrow e^{4t} v = -2e^{4t} + C$$

$$\Rightarrow v = -2 + e^{-4t}$$

$$v = \frac{1}{y^4}$$

$$\Rightarrow \frac{1}{y^4} = e^{-4t} - 2$$

$$\Rightarrow y(t) = \frac{1}{(e^{-4t} - 2)^{\frac{1}{4}}}$$

Solution

$$Q. y' = a_0 y + b_0 y^u$$

$$\Rightarrow \frac{y'}{y^u} = \frac{a_0 y}{y^u} + b_0$$

$$\Rightarrow \frac{y'}{y^u} = \frac{a_0}{y^{u-1}} + b_0 \quad \text{--- (i)}$$

$$\text{Let, } v = \frac{y'}{y^u}.$$

$$\Rightarrow v' = (y^u)'$$

$$\Rightarrow v' = -u \left( \frac{y'}{y^{(u-1)}} \right)$$

$$\Rightarrow -\frac{v'}{u} = \frac{y'}{y^{(u-1)}} \quad \text{--- (ii)}$$

2. From (i) and (ii)  $\Rightarrow$

1.4: Exact Differential Equations:

Any exact differential equation for  $y$  is:

$$N(t, y)y' + M(t, y) = 0$$

where the functions  $N$  and  $M$  satisfy

$$\frac{\partial}{\partial t} N(t, y) = \frac{\partial}{\partial y} M(t, y).$$

Solving exact D.E:

The expression

$$M(x, y)dx + N(x, y)dy \quad \text{--- (i)}$$

is called exact differential if there exists a function  $f(x, y)$  of two real variable such that, the expression equals the total derivative of:

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy.$$

If (i) is exact then, from the D.E, we get:

$$M(x, y) = \frac{\partial f}{\partial x} \quad \text{and} \quad N(x, y) = \frac{\partial f}{\partial y}.$$

Thus, it will be called an exact equation.

Solving exact equation:

$$N(t,y)y' + M(t,y) = 0$$

$$(\partial_t N(t,y) = \partial_y M(t,y)) \rightarrow \text{if it satisfied the exact eqn}$$

$$\partial_y \Psi(t,y) = N(t,y)$$

$$\partial_t \Psi(t,y) = M(t,y)$$

$$0 = N(t,y)y' + M(t,y)$$

$$= \partial_y \Psi(t,y)y' + \partial_t \Psi(t,y)$$

$$= \frac{d}{dt} \Psi(t,y)$$

$$= \int \frac{d}{dt} \Psi(t,y) \rightarrow \Psi(t,y) = C$$

Q. Show that the DE is exact or not?

$$2tyy' + 2t + y^2 = 0$$

$$(2ty)y' + (2t + y^2) = 0$$

$$N(t,y) = 2ty$$

$$M = 2t + y^2$$

$$\partial_t N(t,y) = 2y$$

$$\partial_y M(t,y) = 2y$$

$$\therefore \partial_t N(t,y) = \partial_y M(t,y) \rightarrow \text{exact solution}$$

$$Q. (\sin t)y' + t^2 e^y y' - y' = -y \cos(t) - 2te^y$$

Is it exact/not?

Solution:  $(\sin t + t^2 e^y - 1)y' + y \cos(t) + 2te^y = 0.$

$$N(t, y) = \sin t + t^2 e^y - 1.$$

$$M(t, y) = y \cos t + 2te^y$$

$$\frac{\partial}{\partial y} N(t, y) = \cos t + 2te^y$$

$$\frac{\partial}{\partial t} M(t, y) = \cos t + 2te^y$$

$\therefore$  It is an exact equation.

SHOWN.



1.4.7: Find the sol<sup>n</sup>  $y$  to the eqn:

$$\sin(t)y' + t^2 e^y y' - y' + y \cos t + 2te^y - 3t^2 = 0$$

$$\Rightarrow (\sin t + t^2 e^y - 1)y' + (y \cos t + 2te^y - 3t^2) = 0$$

$$N(t,y) = \sin t + t^2 e^y - 1$$

$$\partial_t N(t,y) = \cos t + 2te^y$$

$$M(t,y) = y \cos t + 2te^y - 3t^2$$

$$\partial_y M(t,y) = \cos t + 2te^y$$

} equal

According to Poincaré's theorem,  
there is a potential function  
that satisfies  $\Psi(t,y)$

$$\partial_y \Psi(t,y) = N(t,y)$$

$$\partial_t \Psi(t,y) = M(t,y)$$

$$\int \frac{\partial}{\partial y} \Psi(t, y) dy = \int N(t, y) dy + g(t)$$

$N$  ~~is~~

$$(\Psi(t, 1) = c.$$

$$\Rightarrow \Psi(t, y) = \int (\sin t + t^2 e^y - 1) dy + g(t).$$

$$\Rightarrow \Psi(t, y) = y \sin t + t^2 e^y - y + g'(t).$$

$$\frac{\partial}{\partial t} \Psi(t, y) = M(t, y)$$

$$\frac{\partial}{\partial t} \Psi(t, y) = y \cos t + 2te^y + g'(t) \text{ --- (i)}$$

$$M(t, y) = y \cos t + 2te^y - 3t^2 \text{ --- (ii)}$$

From (i) and (ii)  $\Rightarrow$

$$g'(t) = -3t^2$$

$$\Rightarrow \int g'(t) dt = \int -3t^2 dt$$

$$\Rightarrow g(t) = -t^3 + c.$$

if  $c = 0$ :

$$g(t) = -t^3$$

*Ans*

Solving exact DE:

1. Check exact/not
2.  $N(t, y) = \frac{\partial}{\partial y} \Psi(t, y)$ ,  
 $M(t, y) = \frac{\partial}{\partial t} \Psi(t, y)$
3. Integrate.
4. Find values of constant.
5.  $\Psi(x, y) = c.$

Subject .....

Date .....

Time .....

MATH-4241:Chapter-4 (Pss)\* Explicit Methods of higher order, <sup>linear</sup> differential eq<sup>n</sup>:

$$a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + \dots + a_n y = f_1(x).$$

Solution,

$$\psi(t, y) = c$$

$$\psi(t, y) = y \sin t + t^2 e^y - y = t^3.$$

$$\text{So, } y \sin t + t^2 e^y - y - t^3 = c.$$

\* Semi-exact equation:

A semi-exact DE is a non-exact equation that can be transformed into an exact equation by multiplying with an integrating factor.

Example:  $(t^2 + ty)y' + 3ty + y^2 = 0$ .

$$\begin{array}{l|l} N(t,y) = t^2 + ty & M(t,y) = 3ty + y^2 \\ \Rightarrow \partial_t N(t,y) = 2t + y & \Rightarrow \partial_y M(t,y) = 3t + 2y \end{array}$$

Not exact, Semi-exact DE.

$Ny' + M = 0 \rightarrow$  not exact.

$\Rightarrow \mu Ny' + \mu M = 0 \quad \text{--- (i)}$

where  $\mu$  has to satisfy,

$\partial_t(\mu N) = \partial_y(\mu M) \rightarrow \mu$  depends on  $t$  only.

$y' = a(t)y + b(t)$

$\mu = e^{-A(t)}$

$A(t) = \int a(t) dt$



1. findings

$$S_t(\mu N) = S_y(\mu M)$$

$$\Rightarrow \mu' N + \mu S_t N = \mu S_y M$$

$$\Rightarrow \mu' N = \mu S_y M - \mu S_t N$$

$$\Rightarrow \mu' = \frac{\mu S_y M - \mu S_t N}{N}$$

$$\therefore \mu' = \frac{(S_y M - S_t N) \mu}{N} \quad \text{--- (ii)}$$

here, 
$$h(t) = \frac{S_y M - S_t N}{N}$$

→ Condition for  
semi exact eqn

In (ii)  $\Rightarrow$

$$\mu' = h(t) \mu$$

$$\Rightarrow \frac{\mu'}{\mu} = h(t)$$

$$\Rightarrow \frac{d}{dt} \ln(\mu(t)) = h(t)$$

$$\Rightarrow \ln(\mu(t)) = H(t)$$

$$\Rightarrow \mu(t) = e^{H(t)} \cdot e^{c_0}$$

$$\Rightarrow \mu(t) = e^{H(t)}$$

In ①  $\Rightarrow$

$$e^H N y' + e^H M = 0.$$

$$h(t) = \frac{3t + 2y - 2t - y}{t^2 + ty}$$

$$= \frac{t+y}{t(t+y)}$$

$$= \frac{1}{t}.$$

$$\begin{aligned} \mu &= e^H \\ &= e^{\int \frac{1}{t} dt} \\ &= e^{\ln t} \\ &= t \end{aligned}$$

$$e^H N(t, y) y' + e^H M = 0.$$

$$\Rightarrow t(t^2 + ty) y' + t(3t + y^2) = 0$$

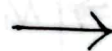
$$\Rightarrow \underbrace{(t^3 + t^2 y) y'}_N + \underbrace{(3t^2 + y^2 t)}_M = 0$$

$$\left. \begin{aligned} S_t N &= 3t^2 + 2ty \\ S_y M &= 3t^2 + 2ty \end{aligned} \right\} \text{exact equation } \underline{\underline{\text{NOW!}}}$$

The equation for inverse function:

If  $Ny' + M$  is exact, then

$$\frac{Mx' + N}{(y-1)'} \text{ is exact.}$$



①  $t \rightarrow x$

$$N(x, y)y' + M(x, y) = 0$$

$$y = y(x) \\ \Rightarrow y' = \frac{dy}{dx}$$

②  $x(y) = x_1$   
 $\Rightarrow y(x_1) = y_1$  [denoting  $x(y)$  as inverse of  $y(x)$ ]

③  $x'(y) = \frac{1}{y'(x)}$  [relation of derivatives between a function and its inverse function]

$$Ny' + M = 0 \longrightarrow S_x N = S_y M$$

Dividing with  $y'$ :

$$S_y M(x, y) = S_x N(x, y)$$

$$\frac{Ny'}{y'} + \frac{M}{y'} = 0$$

$$\Rightarrow N + M(y^{-1})' = 0 \quad [x'(y) = \frac{1}{y'(x)}]$$

$$\Rightarrow N + M_x = 0$$

$$\Rightarrow \boxed{Mx' + N = 0}$$

### \* Applications of DE:-

1. Exponential Decay.
2. Newton's cooling law.
3. Mixing Material.

\* Thm: The solution  $N$  of the exponential decay equation  $N' = -kN$  and initial constant  $N(0) = N_0$  is

$$N(t) = N_0 e^{-kt}$$

Proof: Solving integrating factor:  $e^{kt}$

$$N' = -kN$$

$$\Rightarrow N' + kN = 0$$

$$\Rightarrow (N' + kN) e^{kt} = 0$$

$$\Rightarrow \int (e^{kt} N)' = \int 0$$

$$\Rightarrow e^{kt} N = c$$

$$\Rightarrow N = c e^{-kt}.$$

$$\Rightarrow N = N_0 e^{-kt}.$$

Thm: A radioactive material constant  $k$  and half life  $\tau$  are related by the equation:

$$k\tau = \ln 2$$

Proof:

$$N(t) = N_0 e^{-kt} \quad \text{--- (i)}$$

$$N(t) = \frac{N_0}{2} \quad \text{--- (ii)}$$

$$\text{(i) \& (ii)} \Rightarrow \frac{N_0}{2} = N_0 e^{-kt}$$

$$\Rightarrow \frac{1}{2} = e^{-kt}$$

$$\Rightarrow \ln\left(\frac{1}{2}\right) = -k\tau$$

$$\Rightarrow k\tau = \ln 2$$

$$\Rightarrow k = \frac{\ln 2}{\tau}$$

$$\begin{aligned} N(t) &= N_0 e^{-\frac{1}{\tau} \ln 2 \cdot t} \\ &= N_0 e^{\ln [2^{-t/\tau}]} \\ &= N_0 2^{-t/\tau} \end{aligned}$$

$$\text{If } \tau = t$$

$$N(t) = N_0/2$$

1.5.1:

$$N(t) = \frac{14}{100} N(0)$$

$$N(t) = N(0) 2^{-t/\tau}$$

$$\text{where } \tau = 5730.$$



\* Newton's cooling of material:

$$(\Delta T)' = -k(\Delta T)$$

and the solution of this equation,

$$T(t) = (T_0 - T_s)e^{-kt} + T_s.$$

1.5.2:

A.T.Q,  $T_0 = 45^\circ\text{C}$ ,  $T_s = 5^\circ\text{C}$

$$T(t) = (45 - 5)e^{-kt} + 5$$

$$\Rightarrow T(t) = 40e^{-kt} + 5$$

$$T(2) = 25^\circ; t = 2$$

$$\Rightarrow 25 = 40e^{-kt} + 5$$

$$\Rightarrow 25 - 5 = 40e^{-kt}$$

$$\Rightarrow 20 = 40e^{-kt}$$

$$\Rightarrow \frac{1}{2} = e^{-kt}$$

$$\Rightarrow \frac{1}{2} = e^{-2k}$$

$$\Rightarrow \ln \frac{1}{2} = -2k$$

$$\Rightarrow k = \frac{1}{2} \ln 2$$

$$\therefore k = \ln \sqrt{2}$$

MATH-4241:Chapter-4 (Pss)

linear

\* Explicit Methods of higher order differential eq<sup>n</sup>:

on-homogenous:  $a_0 \frac{d^ny}{dx^n} + a_1 \frac{d^{n-1}y}{dx^{n-1}} + a_2 \frac{d^{n-2}y}{dx^{n-2}} + \dots + a_n y = f_1(x).$

homogenous:  $a_0 \frac{d^ny}{dx^n} + a_1 \frac{d^{n-1}y}{dx^{n-1}} + a_2 \frac{d^{n-2}y}{dx^{n-2}} + \dots + a_n y = 0 \quad \text{--- (i)}$

$y = e^{mx} \rightarrow$  trial solution.

$$\Rightarrow \frac{dy}{dx} = m e^{mx}$$

$$\Rightarrow \frac{d^2y}{dx^2} = m^2 e^{mx}$$

⋮

$$\Rightarrow \frac{d^ny}{dx^n} = m^n e^{mx}.$$

$$\Rightarrow a_0 m^n e^{mx} + a_1 m^{n-1} e^{mx} + \dots + a_n e^{mx} = 0$$

$$\Rightarrow e^{mx} (a_0 m^n + a_1 m^{n-1} + a_2 m^{n-2} + \dots + a_n) = 0$$

$$\Rightarrow \boxed{a_0 m^n + a_1 m^{n-1} + a_2 m^{n-2} + \dots + a_n = 0} \rightarrow \text{Auxiliary Equation}$$

$[as \ e^{mx} \neq 0]$

$\rightarrow$  roots:  $m = x_1, x_2, x_3, \dots$

# General Solution:

$$y = c_1 e^{x_1 x} + c_2 e^{x_2 x} + \dots + c_n e^{x_n x}$$

# Distinct real roots of  $\eta$ :

$$\# (\mathcal{D}^2 + 3\mathcal{D} + 2)y = 0$$

$$\Rightarrow \frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} + 2y = 0$$

~~For~~  $y = e^{mx}$  might be the trial soln

$$\frac{dy}{dx} = m e^{mx}, \quad \frac{d^2 y}{dx^2} = m^2 e^{mx}$$

$$\Rightarrow m^2 e^{mx} + 3m e^{mx} + 2e^{mx} = 0$$

$$\Rightarrow e^{mx} (m^2 + 3m + 2) = 0$$

$$\Rightarrow \boxed{m^2 + 3m + 2} \longrightarrow \text{Auxiliary Equation}$$

$$\Rightarrow (m+2)(m+1) = 0$$

$$\therefore m = -1, -2. \text{ (independent)}$$

$$\text{G.S} \Rightarrow y = c_1 e^{-2x} + c_2 e^{-x}$$

Ans.

G.S:  $y = (c_1 + c_2x + c_3x^2 + \dots + c_n x^{n-1}) e^{mx}$   
 (for  $(n-1)$  repeated real roots)

# Repeated real roots of  $m$ :

$$\Rightarrow \frac{d^2y}{dx^2} - 6 \frac{dy}{dx} + 9y = 0$$

$$\Rightarrow m^2 - 6m + 9 = 0$$

$$\Rightarrow (m - 3)^2 = 0$$

$$\therefore m = 3, 3.$$

G.S:  $y = c_1 e^{3x} + c_2 e^{3x} \cdot x$   
 $= e^{3x} (c_1 + c_2 x)$

#  $(k-1)$  repeated real roots among  $n$  and others are not.

#  $y = (c_1 + c_2x + c_3x^2 + \dots + c_k x^{k-1}) e^{mx}$   
 $+ c_{k+1} e^{m_1x} + c_{k+2} e^{m_2x} + \dots + c_n e^{m_nx}.$

$$\# \frac{d^4y}{dx^4} - 5 \frac{d^3y}{dx^3} + 6 \frac{d^2y}{dx^2} + 4 \frac{dy}{dx} - 8y = 0$$

$$\Rightarrow m^4 - 5m^3 + 6m^2 + 4m - 8 = 0$$

$$\therefore m = 2, 2, 2, -1$$

G.S:  $y = (c_1 + c_2x + c_3x^2) e^{2x} + c_4 e^{-x}$

Ans.

## # Conjugate complex roots of $m$ :

$$m = a \pm ib.$$

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = 0$$

$$\Rightarrow m^2 + m + 1 = 0$$

$$\Rightarrow m = \frac{-1 \pm i\sqrt{3}}{2}$$

$$\therefore a = -1/2, b = \sqrt{3}/2$$

$$y = k_1 e^{m_1 x} + k_2 e^{m_2 x}$$

$$= k_1 e^{(a+ib)x} + k_2 e^{(a-ib)x}$$

$$= k_1 e^{ax} \cdot e^{ibx} + k_2 e^{ax} \cdot e^{-ibx} \quad [e^{i\theta} = \cos\theta + i\sin\theta]$$

$$= e^{ax} [k_1 (\cos bx + i\sin bx) + k_2 (\cos bx - i\sin bx)]$$

$$= e^{ax} [(k_1 + k_2) \cos bx + i(k_1 - k_2) \sin bx]$$

$$= e^{ax} (c_1 \cos bx + c_2 \sin bx)$$

$$\# \frac{(a+ib)}{k}, \frac{(a-ib)}{k} \rightarrow k \text{ fold roots.}$$

$$y = e^{ax} \left[ (c_1 + c_2x + c_3x^2 + \dots + c_kx^{k-1}) \sin bx + (c_{k+1} + c_{k+2}x + c_{k+3}x^2 + \dots + c_{2k}x^{k-1}) \cos bx \right]$$

$$\# \frac{d^4y}{dx^4} - 4 \frac{d^3y}{dx^3} + 14 \frac{d^2y}{dx^2} - 20 \frac{dy}{dx} + 25y = 0$$

$$\Rightarrow m^4 - 4m^3 + 14m^2 - 20m + 25 = 0$$

$$\therefore m = (1 \pm 2i), (1 \pm 2i)$$

$$y = e^x [(c_1 + c_2x) \sin 2x + (c_3 + c_4x) \cos 2x] \quad A$$

**PDE**  $\rightarrow$  more than one independent variable.

for solving,

$\rightarrow$  Lagrange's Method

$\rightarrow$  some standard forms. (1, 2, 3, 4).

$\rightarrow$  Charpit's method.

**Lagrange's Method:**

Equation of the type  **$Pp + Qq = R$**  ——— (i)  
is called Lagrange's PDE where (capital)  
 $P, Q, R$  are the functions of  $x, y, z$ .

The auxiliary equation is  **$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$**

where;  $p = \frac{\partial z}{\partial x}$ ,  $q = \frac{\partial z}{\partial y}$ , ~~and~~ ~~for~~

and the solution,

**$f(u, v) = 0$**  where  $u(x, y, z) = c_1$ ,  
 $v(x, y, z) = c_2$

$\rightarrow$  arbitrary function.

$$\# \left( \frac{y^2 z}{x} \right) p + \frac{xz}{y} q = y^2$$

$\downarrow$   
 $P$

$\downarrow$   
 $Q$

$\downarrow$   
 $R$

$$\Rightarrow \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\Rightarrow \frac{dx}{\frac{y^2 z}{x}} = \frac{dy}{xz} = \frac{dz}{y^2} \quad [L.A.E]$$

Taking 1<sup>st</sup> and 3<sup>rd</sup> ratios:

$$\frac{dx}{\frac{y^2 z}{x}} = \frac{dz}{y^2}$$

$$\Rightarrow \int x dx = \int z dz$$

$$\Rightarrow \frac{x^2}{2} - \frac{z^2}{2} = c_1$$

$$\Rightarrow x^2 - z^2 = 2c_1 = c_1$$

Taking 1<sup>st</sup> and 2<sup>nd</sup> ratios:

$$\frac{dx}{\frac{y^2 z}{x}} = \frac{dy}{xz}$$

$$\Rightarrow \int x^2 dx = \int y^2 dy$$

$$\Rightarrow \frac{x^3}{3} - \frac{y^3}{3} = c_2$$

$$\Rightarrow x^3 - y^3 = 3c_2 = c_2$$

$$\therefore f = (x^2 - z^2, x^3 - y^3) = 0$$

Ans

$$\# p + 3q = 5z + \tan(y - 3x).$$

$$P = 1, Q = 3, R = 5z + \tan(y - 3x).$$

$$\Rightarrow \frac{dx}{1} = \frac{dy}{3} = \frac{dz}{5z + \tan(y - 3x)}$$

Taking 1<sup>st</sup> and 2<sup>nd</sup> ratios:

$$\frac{dx}{1} = \frac{dy}{3}$$

$$\Rightarrow \int 3dx = \int dy$$

$$\Rightarrow 3x = y + c_1$$

$$\Rightarrow y - 3x = c_1$$

$$\therefore f(y - 3x, \frac{e^{5x}}{5z + \tan c_1}) = 0$$

Taking 1<sup>st</sup> and 3<sup>rd</sup> ratios:

$$\int \frac{dx}{1} = \int \frac{dz}{5z + \tan(y - 3x)}$$

$$\Rightarrow \int \frac{dx}{1} = \int \frac{dz}{5z + \tan c_1}$$

$$\Rightarrow x = \frac{1}{5} \log_e (5z + \tan c_1) + \log c_2$$

$$\Rightarrow 5x = \log (5z + \tan c_1) + c_2$$

$$\Rightarrow e^{5x} = (5z + \tan c_1) c_2$$

$$\Rightarrow \frac{e^{5x}}{5z + \tan c_1} = c_2$$

Ans.

## Charpit's Method:

$$\# f(x, y, z, p, q) = 0$$

$$\begin{aligned} \text{CAE} \Rightarrow \frac{dp}{\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z}} &= \frac{dq}{\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z}} = \frac{dz}{-p \frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q}} \\ &= \frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}} = \frac{\partial f}{0} \end{aligned}$$

$$\# px + qy = pq \quad \text{--- (1)}$$

$$f(x, y, z, p, q) = px + qy - pq = 0.$$

$$\frac{\partial f}{\partial x} = p, \quad \frac{\partial f}{\partial y} = q, \quad \frac{\partial f}{\partial z} = 0,$$

$$\frac{\partial f}{\partial p} = x - q, \quad \frac{\partial f}{\partial q} = y - p.$$

STEP-01:

Putting these values in C.A.E  $\Rightarrow$

$$\frac{dp}{p+p \cdot 0} = \frac{dq}{q+q \cdot 0} = \frac{dz}{-p(n-q) - q(y-p)} = \frac{dx}{-(n-q)} = \frac{dy}{-(y-p)}$$

**STEP-2:** Taking 1<sup>st</sup> and 2<sup>nd</sup> ratio:

$$\frac{dp}{p} = \frac{dq}{q} \Rightarrow \frac{dp}{p} - \frac{dq}{q} = 0.$$

$$\Rightarrow \ln p - \ln q = \ln a.$$

$$\Rightarrow \ln \frac{p}{q} = \ln a$$

$$\Rightarrow p = aq \quad \text{--- (ii)}$$

Putting (ii) in (i)  $\Rightarrow$

$$aqx + qy = aq^2$$

$$\Rightarrow ax + y = aq.$$

$$\Rightarrow q = \frac{ax+y}{a}$$

Then,  $p = a \cdot \frac{ax+y}{a}$  (in ii)

$$\Rightarrow p = ax+y.$$

**STEP-3:**

$$dz = p dx + q dy.$$

$$\Rightarrow dz = (ax+y) dx + \left(\frac{ax+y}{a}\right) dy.$$

$$\Rightarrow a dz = (ax+y) \cdot a dx + (ax+y) dy.$$

$$\Rightarrow a dz = (ax+y) (a dx + dy).$$

$$\Rightarrow \int a dz = \int \underbrace{(ax+y)}_u \underbrace{d(ax+y)}_{dx}.$$

$$\Rightarrow az = \frac{(ax+y)^2}{2} + c$$

*Ans*